Kinship Structures (3)

Extensions of the kinship model

Diego Alburez Gutiérrez[†]

[†]Kinship Inequalities Research Group,
Max Planck Institute for Demographic Research

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Agenda

1. Time-variance
2. Two-sex models
3. Other extensions

Recap

1. What is the general form of the (time-invariant) kinship models?
2. What are the different 'model specifications' that we discussed yesterday?

Typology of kinship models

No	time	sex	state	reference
1	invariant	female	age	
2	variant	female	age	
3	invariant	two	age	
4	invariant	female	multiple	
5	variant	two	multiple	

Time-variance

Typology of kinship models

No	time	sex	state	reference
1	invariant	female	age	
2	**variant**	**female**	**age**	[1]
3	invariant	two	age	
4	invariant	female	multiple	
5	variant	two	multiple	

[1] Caswell, H., & Song, X. (2021). The formal demography of kinship. III. kinship dynamics with time-varying demographic rates. *Demographic Research*, *45*, 517–546

Time-variant kinship models

1. Demographic rates change over time
2. Past demographic change $\rightarrow$ contemporary kinship structures
3. E.g., mortality crises, baby booms
4. Estimates by age, period, and cohort

Recap: Time-invariant, one-sex model

The models are of the general form:

$$\underbrace{\mathbf{k}(x+1)}_{\substack{\text{age structure of kin}\\ \text{at Focal's age } x+1}} = \underbrace{\mathbf{U}\,\mathbf{k}(x)}_{\substack{\text{ageing and survival}\\ \text{of existing kin}}} + \underbrace{\begin{cases} \mathbf{0} \\ \mathbf{F}\,\mathbf{k}^*(x) \end{cases}}_{\substack{\text{new kin members}\\ \text{added to the population}}} .$$

where:

- $\mathbf{U}$ a matrix with survival probabilities in the subdiagonal
- $\mathbf{F}$ a matrix with fertility rates in the first row

Time-variant, one-sex model

The models are of the general form:

$$\underbrace{\mathbf{k}(x+1,t+1)}_{\substack{\text{age structure of kin}\\ \text{at Focal's age } x+1\\ \text{and time } t+1}} = \underbrace{\mathbf{U_t}\,\mathbf{k}(x,t)}_{\substack{\text{ageing and survival}\\ \text{of existing kin}}} + \underbrace{\begin{cases} \mathbf{0} \\ \mathbf{F_t}\,\mathbf{k}^*(x,t) \end{cases}}_{\substack{\text{new kin members}\\ \text{added to the population}}} .$$

where:

- $\mathbf{U_t}$ a matrix with time-variant survival probabilities in the subdiagonal
- $\mathbf{F_t}$ a matrix with time-variant fertility rates in the first row

Boundary conditions

figures/boundary.PNG

Boundary conditions

- specify the complete age vector at time $t = 0$

$$\mathbf{k}(x, 0) \qquad x = 0, ..., \omega.$$

- Specify the initial vector at each time

$$\mathbf{k}(0, t) \qquad t = 0, ..., \omega.$$

Daughters

Daughters ($\mathbf{a}$) are the result of the reproduction of Focal:

$$\underbrace{\mathbf{a}(x+1,t+1)}_{\substack{\text{age structure of daughters}\\ \text{at Focal's age } x+1}} = \underbrace{\mathbf{U_t}\,\mathbf{a}(x,t)}_{\substack{\text{ageing and survival}\\ \text{of existing daughters}}} + \underbrace{\mathbf{F_t}\,\mathbf{e}_x}_{\substack{\text{new daughters}\\ \text{(subsidy)}}}. \tag{1}$$

$$a(0) = \mathbf{0}.$$

where:

- $\mathbf{U_t}$ is a matrix with time-variant survival probabilities in the subdiagonal
- $\mathbf{F_t}$ is a matrix with time-variant fertility rates in the first row
- $\mathbf{F_t}\,\mathbf{e}_x$ is the subsidy vector
- e_x is the unit vector for age x
- $a(0)$ is the distribution of daughters at Focal's birth

Mothers

The population of mothers ($\mathbf{d}$) of Focal consists of at most a single individual:

$$\underbrace{\mathbf{d}(x+1,t+1)}_{\substack{\text{age structure of mothers}\\ \text{at Focal's age } x+1}} = \underbrace{\mathbf{U_t}\,\mathbf{d}(x,t)}_{\substack{\text{ageing and survival}\\ \text{of existing mothers}}} + \underbrace{\mathbf{0}.}_{\substack{\text{new mothers}\\ \text{(subsidy)}}} \tag{2}$$

$$d(0) = \pi(t).$$

where:

- $b(0)$ is the distribution of mothers at Focal's birth
- $\pi(t)$ is the distribution of ages of mothers in the population

Age, period, and cohort in kinship models

figures/cohort.png

Period results for numbers of daughters and granddaughters in Sweden

figures/pe.PNG

Cohort results for numbers of daughters and granddaughters in Sweden

figures/co.PNG

Age results for numbers of daughters and granddaughters in Sweden

figures/age.PNG

Discuss

1. What is the difference between the expected number of daughters calculated using (i) a time-invariant model, and (ii) the period dimension of the time-variant model?
2. What is the difference between the period and cohort results of the time-variant model?
3. How do the time-variant models deal with missing demographic data before a certain year (e.g., the UN only reports data starting in 1950)?

Break

Two-sex models

Typology of kinship models

No	time	sex	state	reference
1	invariant	female	age	
2	variant	female	age	
3	**invariant**	**two**	**age**	[2]
4	invariant	female	multiple	
5	variant	two	multiple	

[2] Caswell, H. (2022). The formal demography of kinship IV: Two-sex models and their approximations. *Demographic Research*, *47*, 359–396.

Why do we need two-sex models?

1. Differential survival and reproduction for women and men
2. These differences are not stable: change over time
3. More pronounced in some settings
4. Estimate male and female kin for male and female Focals

Two-sex model (descendants)

figures/two.PNG

$$\tilde{\mathbf{a}} = \begin{pmatrix} \mathbf{a}'_1 \\ \mathbf{a}'_2 \end{pmatrix}$$

where

Offspring (sons and daughters)

Children ($\tilde{\mathbf{a}}$) are the result of the reproduction of Focal:

$$\underbrace{\tilde{\mathbf{a}}(x+1)}_{\substack{\text{age structure of offspring} \\ \text{at Focal's age } x+1}} = \underbrace{\tilde{\mathbf{U}}\,\tilde{\mathbf{a}}(x)}_{\substack{\text{ageing and survival} \\ \text{of existing offspring}}} + \underbrace{\tilde{\mathbf{F}}\,\tilde{\phi}(x).}_{\substack{\text{new offspring} \\ \text{(subsidy)}}} \tag{3}$$

$$\tilde{a}(0) = \mathbf{0}.$$

where:

- $\tilde{\mathbf{U}}$ is a block-structured matrix of survival probabilities
- $\tilde{\mathbf{F}}$ is a block-structured matrix of fertility rates
- $\tilde{\mathbf{F}}\,\tilde{\phi}(x)$ is the subsidy vector
- $\tilde{\phi}(x)$ is the state vector of a Focal of specified sex
- $\tilde{a}(0)$ is the distribution of offspring at Focal's birth

Blocks-structured input matrices

For mortality:

$$\tilde{\mathbf{U}} = \begin{pmatrix} \mathbf{U}_f & \mathbf{0} \\ \mathbf{0} & \mathbf{U}_m \end{pmatrix}$$

For fertility:

$$\tilde{\mathbf{F}} = \begin{pmatrix} \bar{\alpha}\mathbf{F}_f & \bar{\alpha}\mathbf{F}_m \\ \alpha\mathbf{F}_f & \bar{\alpha}\mathbf{F}_m \end{pmatrix}$$

where

- $\mathbf{U}_f$ is a matrix with female survival probabilities in the subdiagonal
- $\mathbf{F}_f$ is a matrix with female fertility rates in the first row
- α is the proportion males among offspring
- $\bar{\alpha}$ is $1 - \alpha$

Parents

The population of parents ($\tilde{\mathbf{d}}$) of Focal consists of at most a single individual:

$$\underbrace{\tilde{\mathbf{d}}(x+1)}_{\substack{\text{age structure of parents} \\ \text{at Focal's age } x+1}} = \underbrace{\tilde{\mathbf{U}}\,\tilde{\mathbf{d}}(x)}_{\substack{\text{ageing and survival} \\ \text{of existing parents}}} + \underbrace{\mathbf{0}.}_{\substack{\text{new parents} \\ \text{(subsidy)}}} \tag{4}$$

$$\tilde{d}(0) = \tilde{\pi}.$$

where:

- $\tilde{b}(0)$ is the distribution of parents at Focal's birth
- $\tilde{\pi}$ is the distribution of ages of parents in the population

Two-sex kin estimation

figures/tree.PNG

Data requirements

figures/input.PNG

Male and female fertility

figures/fert.PNG

Approximations for two-sex kinship models

1. Androgynous fertility
   - Assume that $\mathbf{F}_m = \mathbf{F}_f$
2. GKP factors
   - Run one-sex model and multiply resulting kinship structure by a 'GKP factor'
   - daughters $\times$ 2, granddaughters $\times$ 4, great-granddaughters $\times$ 8, mothers $\times$ 2, grandmothers $\times$ 4, great-grandmothers $\times$ 8, sisters $\times$ 2, nieces $\times$ 4, aunts $\times$ 4, and cousins $\times$ 8

Expected number of female and male kin in three countries

figures/two-ex.PNG

Discuss

1. Why do we need the 'androgynous' and 'GKP factor' approximations for two-sex kinship models?
2. Which of the two do you think is better? Can you think of other possible 'approximations'?

Other extensions

Typology of kinship models

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5	**variant**	two	**multiple**	[3]

[3] Williams, I., Alburez-Gutierrez, D., & DemoKin Team. (2023). *DemoKin: 1.0.3*. https://CRAN.R-project.org/package=DemoKin

Multistate kinship models

figures/kin_ageedu_gm_bothsex_Nepal.pdf

Kin loss by cause of death[4]

figures/arms.png

Discuss

1. Do you see any application of the kinship models to your own work?
2. Which model specification would be more appropriate for this?